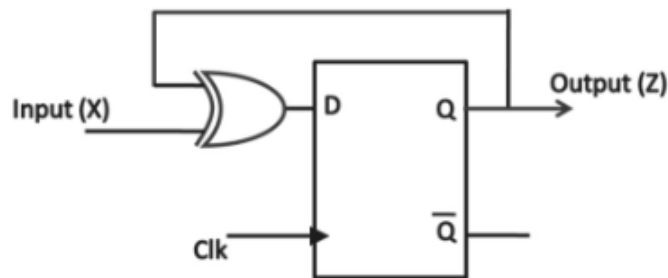


Assignment 4

Sequential- Assignment

1. Explain, the behavior of the given circuit.



2. Design a posedge triggered D flipflop using 2:1 Muxes only.
3. Design a negedge triggered T flipflop using 2:1 Muxes only.
4. Convert by adding external gates:
 - a) A D flip-flop to a J-K flip-flop.
 - b) A T flip-flop to a D flip-flop.
5. In a positive edge triggered JK flip flop, we have $J = Q'$ and $K = 1$ as shown in the figure below. Assume the flip flop was initially cleared and then clocked for 6 clock pulses, find the sequence at the Q output for those 6 clock cycles.

